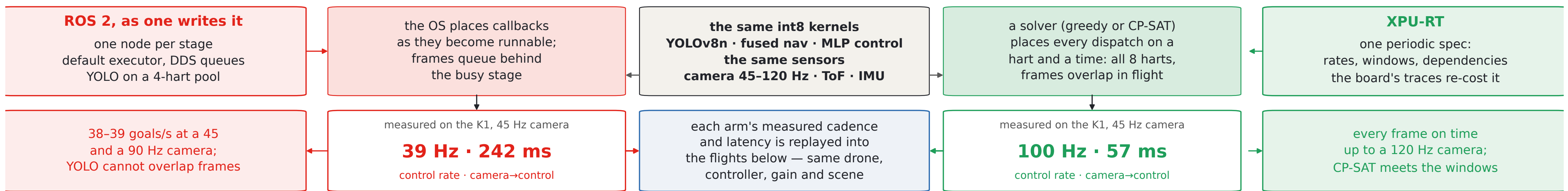
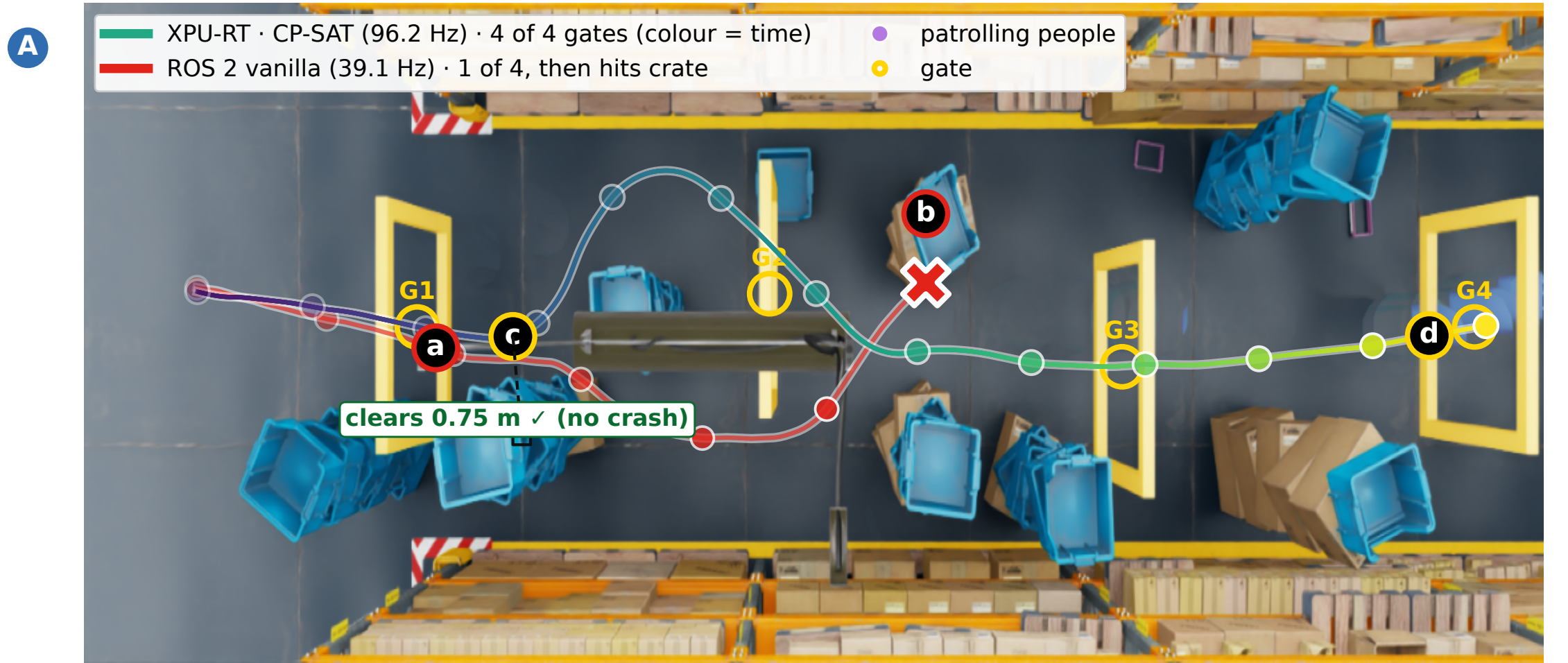


From one flight to the whole programme — the same kernels on one K1, scheduled by XPU-RT or run as a ROS 2 graph; drone, controller, gain and scene are shared, only what the board delivers differs

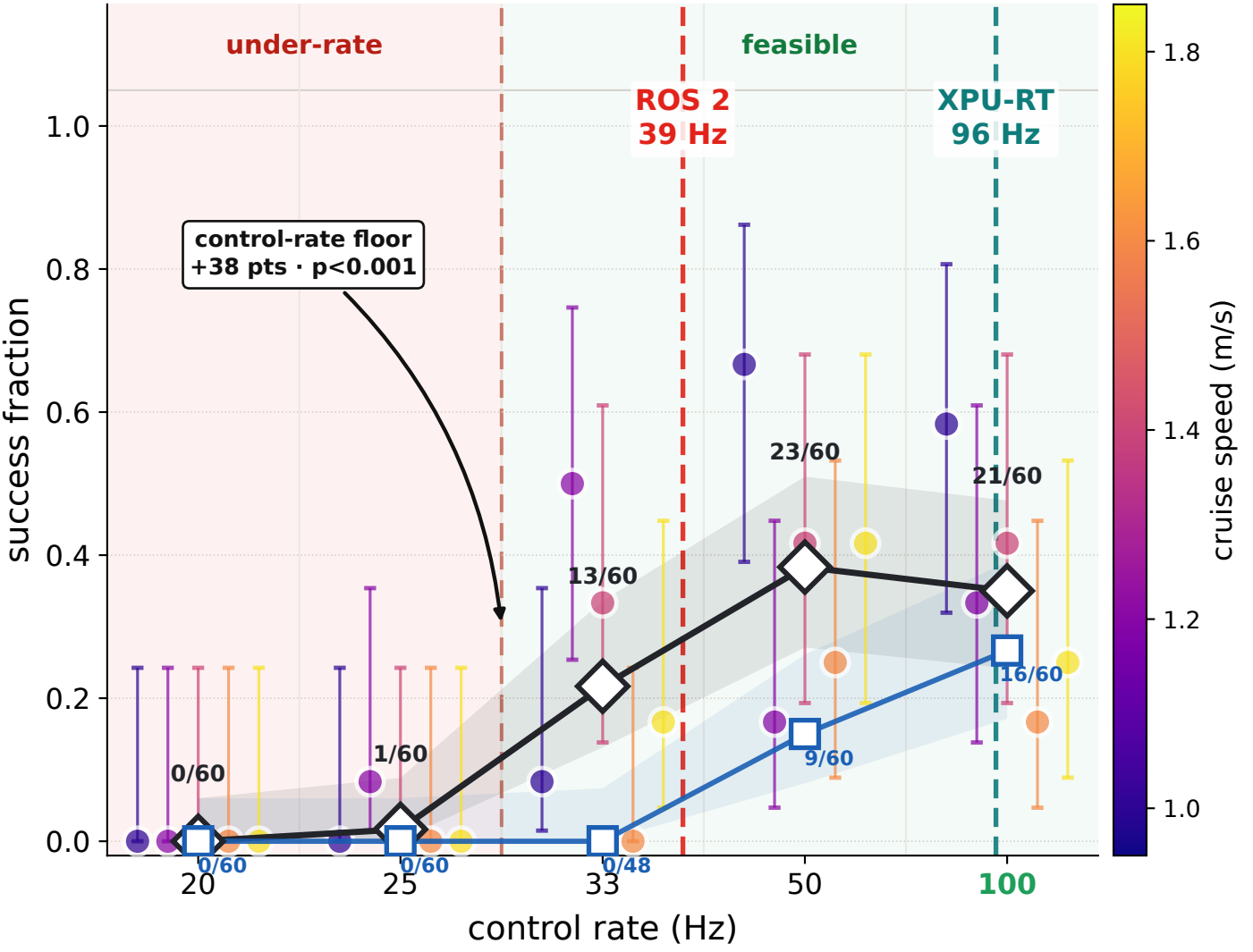
Same kernels, same board, two runtimes — and how their measurements reach the flights below



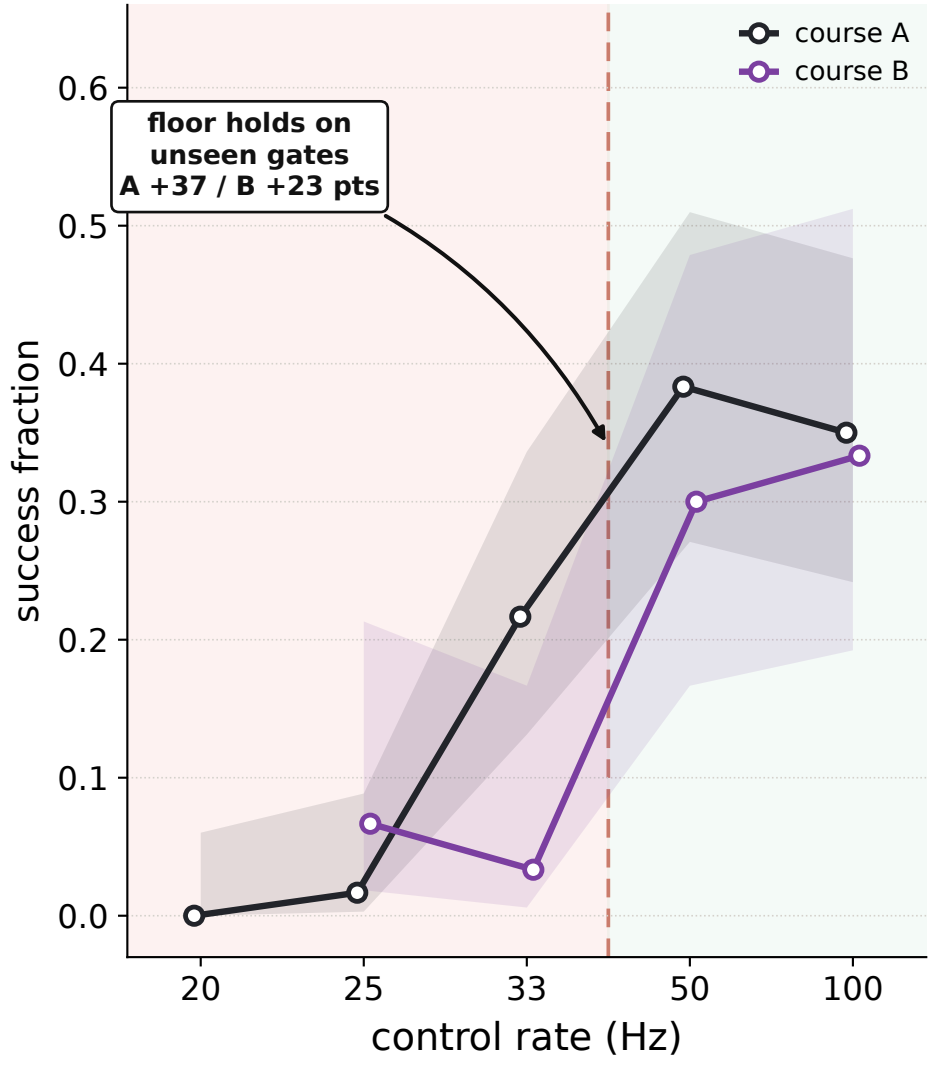
The specific case — one scene, 1.0 m/s: XPU-RT clears the course; ROS 2 clears G1 and hits a crate



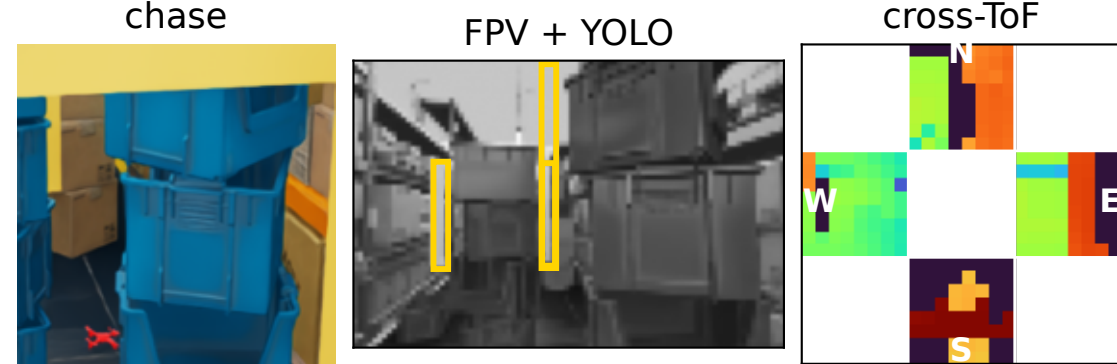
Control-rate envelope (rate injected, 300 flights): success control rate — ROS 2's 39 Hz and XPU-RT's 96 Hz marked



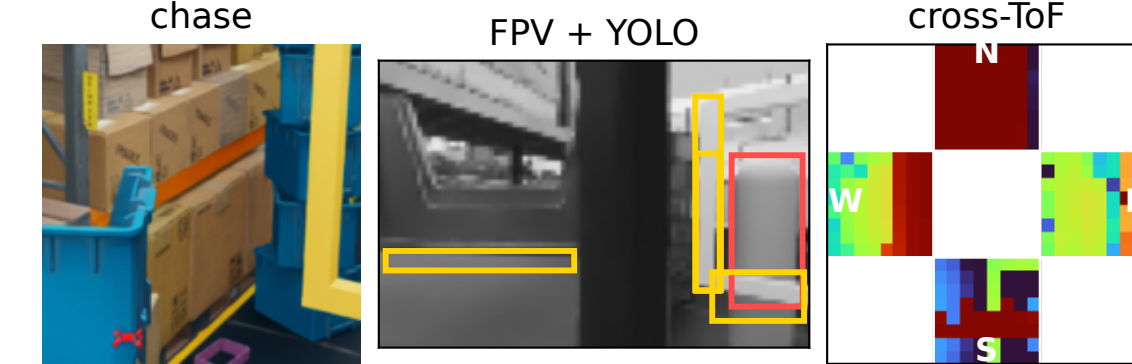
Generalisation: the same floor on an unseen gate course



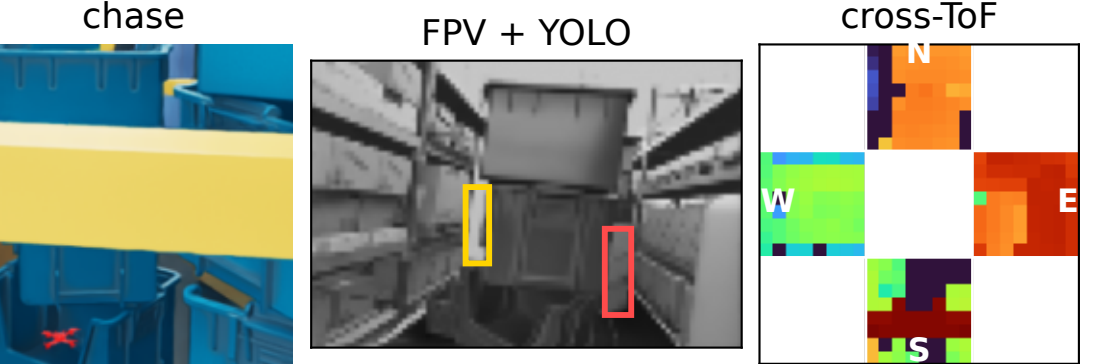
ROS 2 39 Hz · clears G1 · t=3.5s



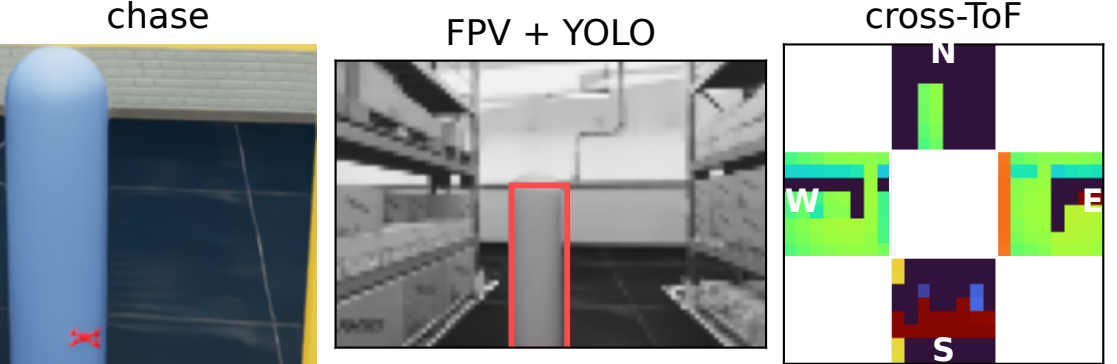
ROS 2 39 Hz · hits crate after G1 · t=9.5s



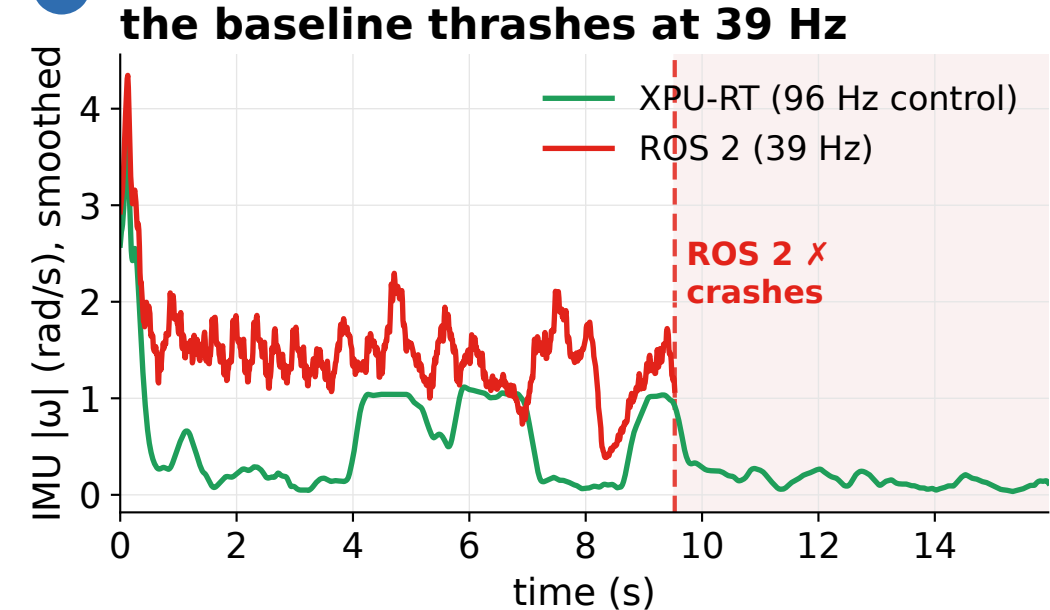
XPU-RT 96 Hz · clears crate 0.75 m · t=4.3s



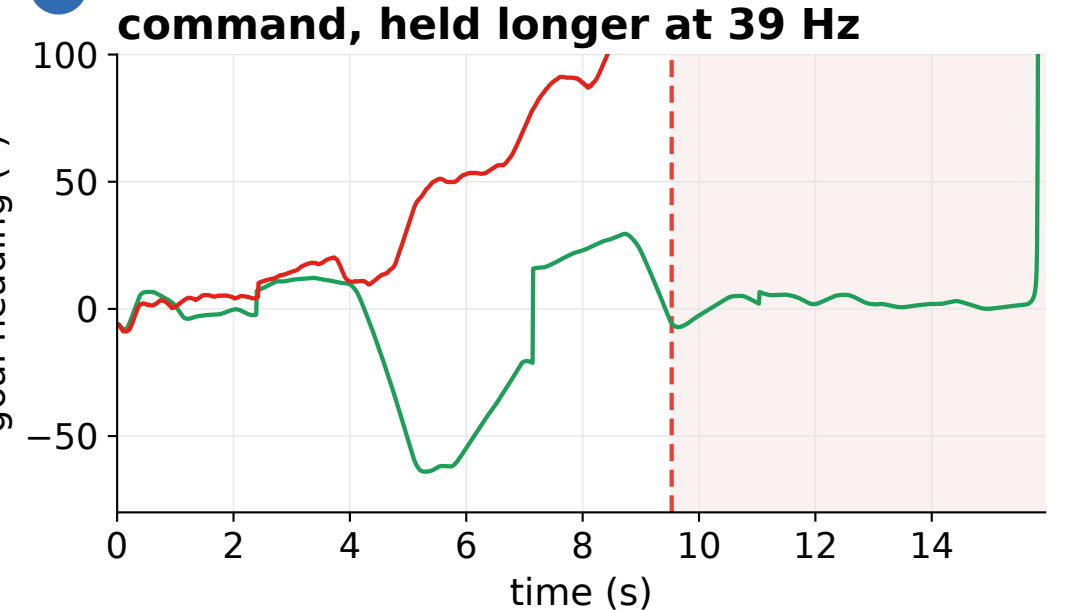
XPU-RT 96 Hz · reaches gate · t=15.3s



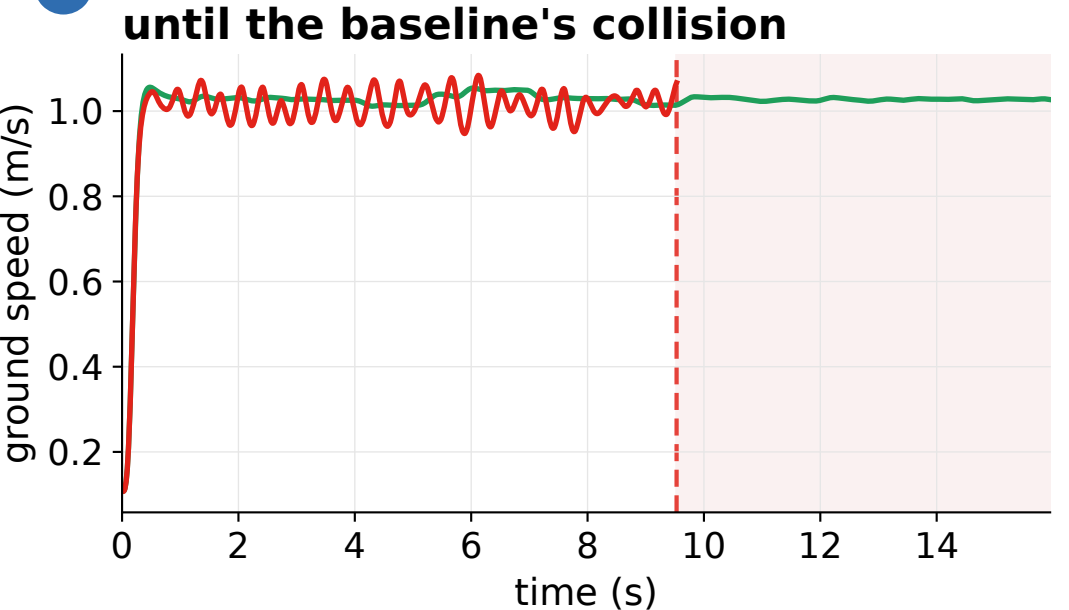
Inside the pair — body rate: the baseline thrashes at 39 Hz



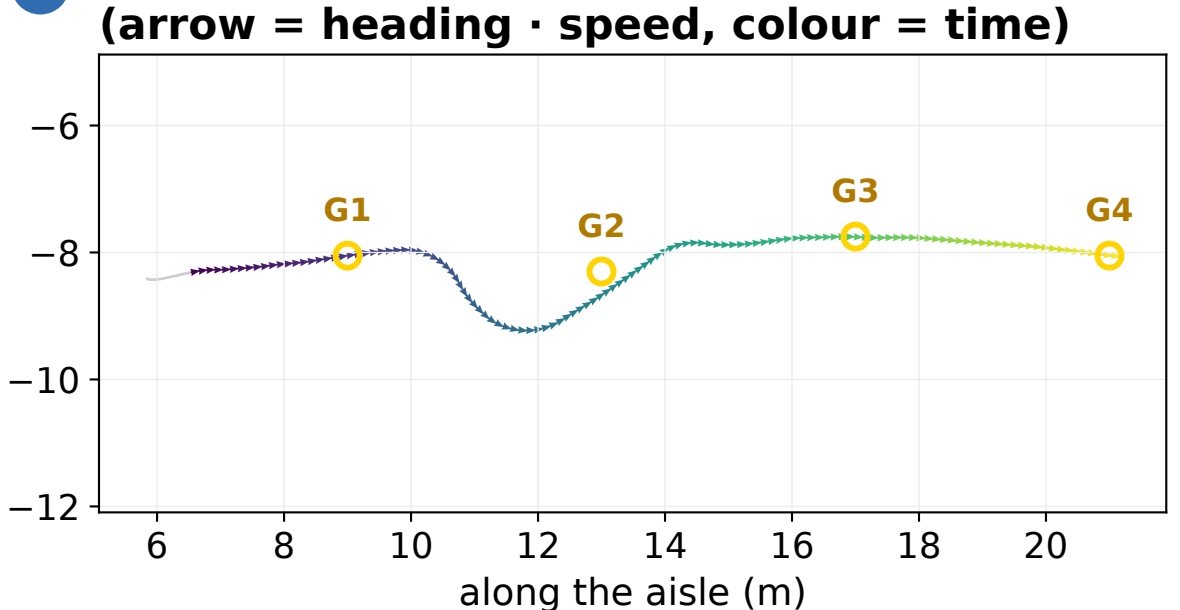
Nav goal heading: the fused nav's command, held longer at 39 Hz



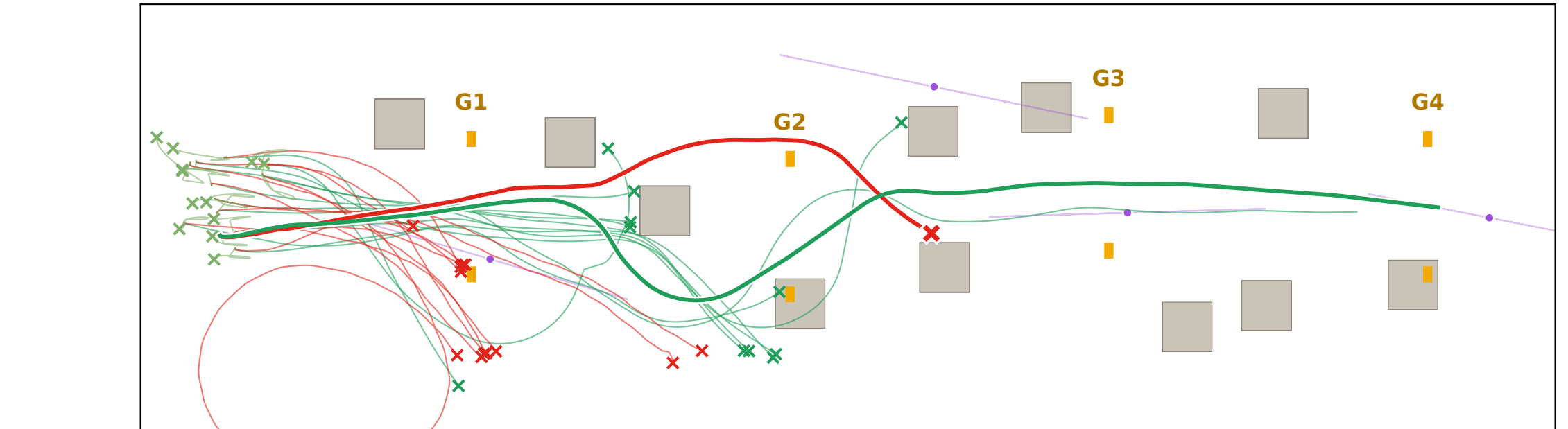
Forward speed: the same cruise until the baseline's collision



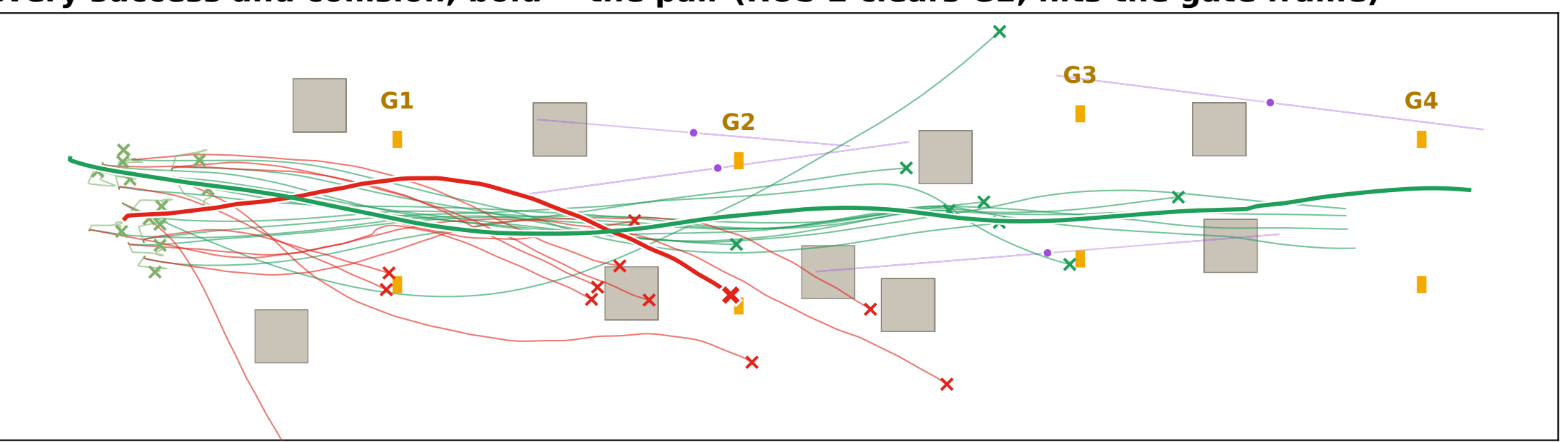
XPU-RT's velocity through the gates (arrow = heading · speed, colour = time)



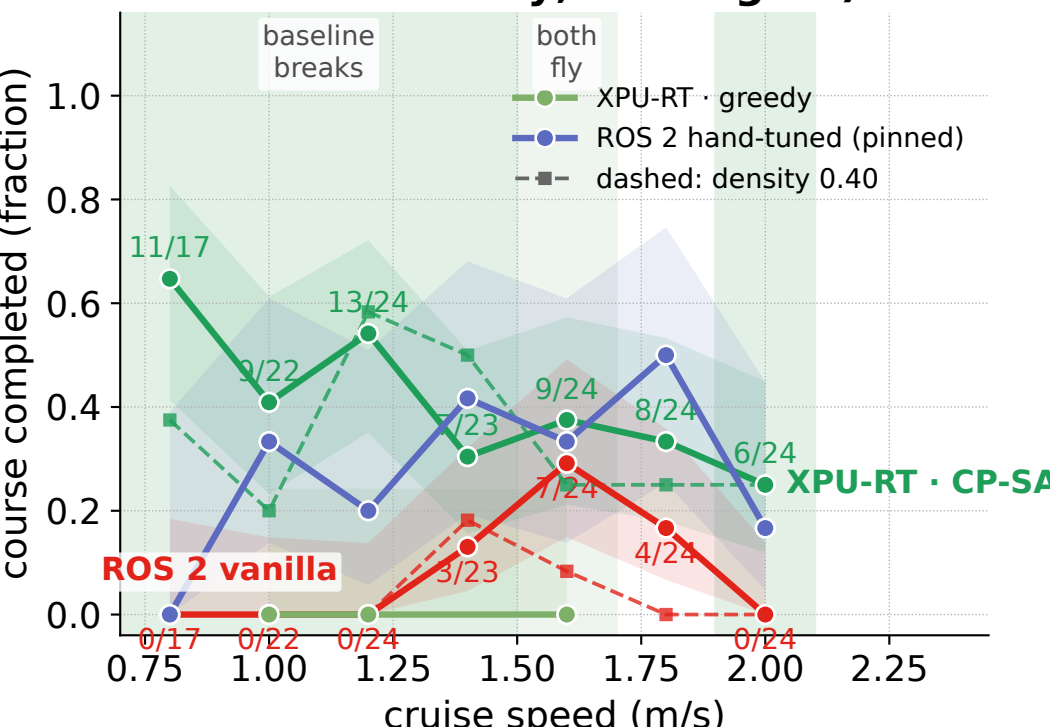
The display scene (1.0 m/s), flown 36 times: CP-SAT 1/12, ROS 2 0/12, greedy 0/12 every success and collision; bold = the pair (ROS 2 clears G1, hits the crate)



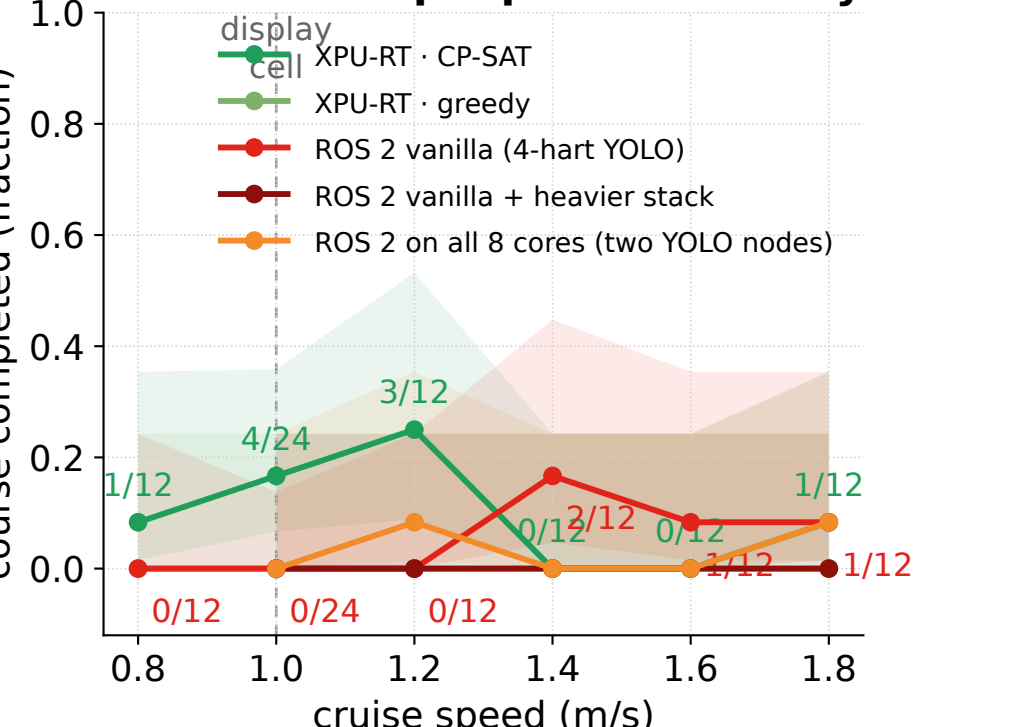
A second scene (1.4 m/s), flown 36 times: CP-SAT 4/12, ROS 2 0/12, greedy 0/12 every success and collision; bold = the pair (ROS 2 clears G2, hits the gate frame)



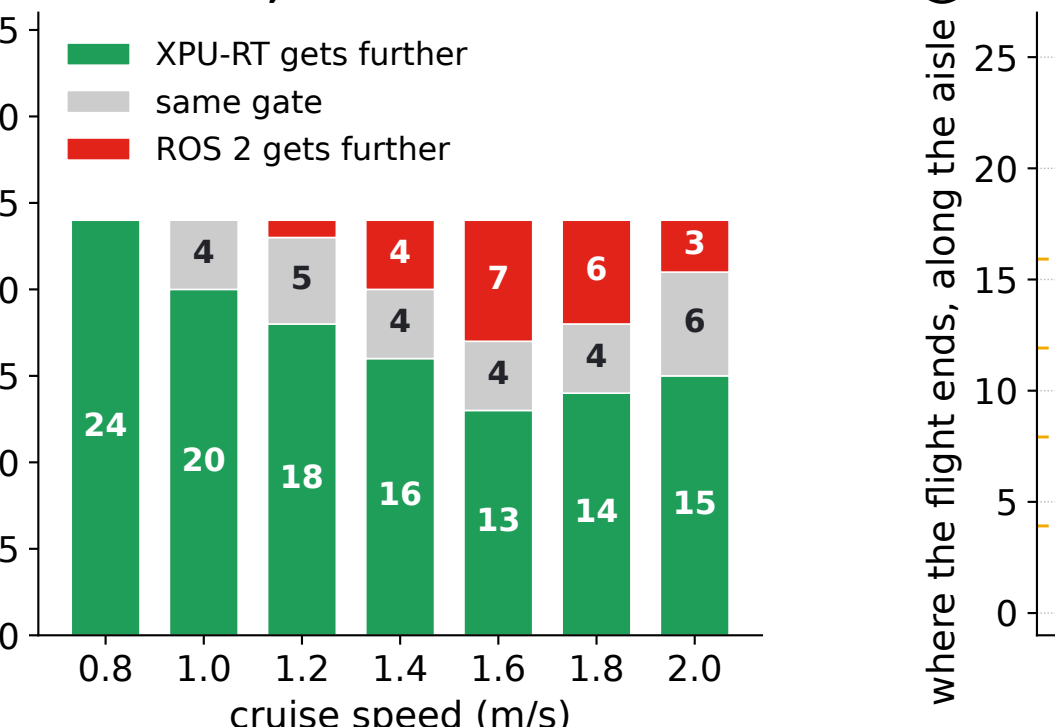
The breaking point (1.7 m people, cadence + latency; 428 flights)



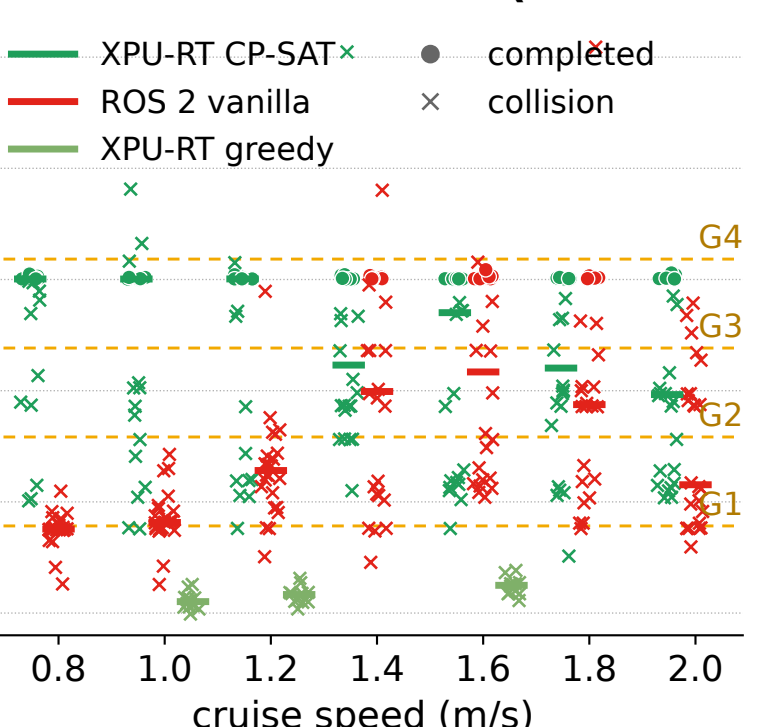
The 2.4 m scene of A: ahead to 1.2 m/s, from 0.8 the people block every arm



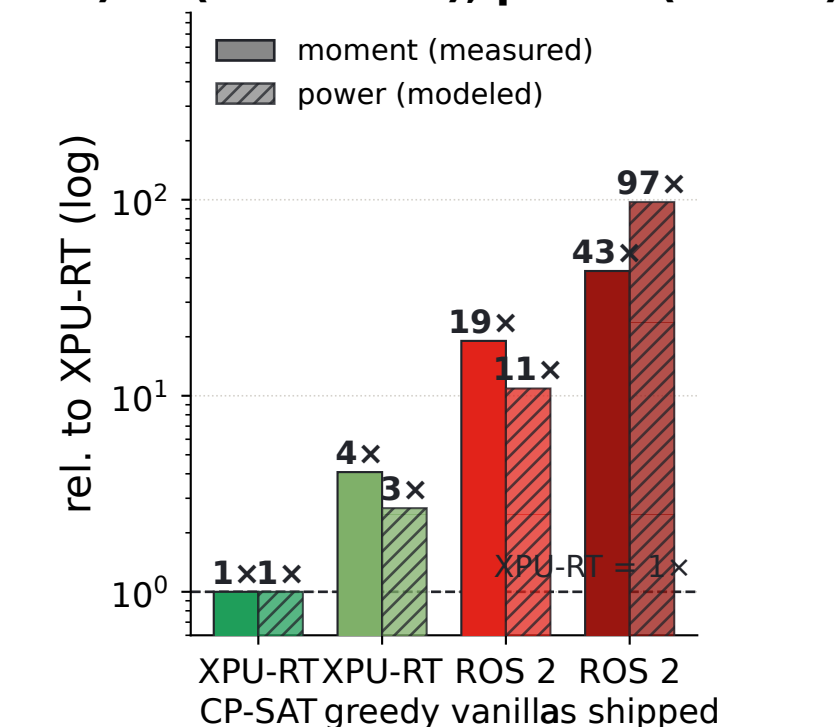
Same seed, 1.7 m: XPU-RT further 120 vs 21, 27 tied · 2.4 m: 49 vs 15



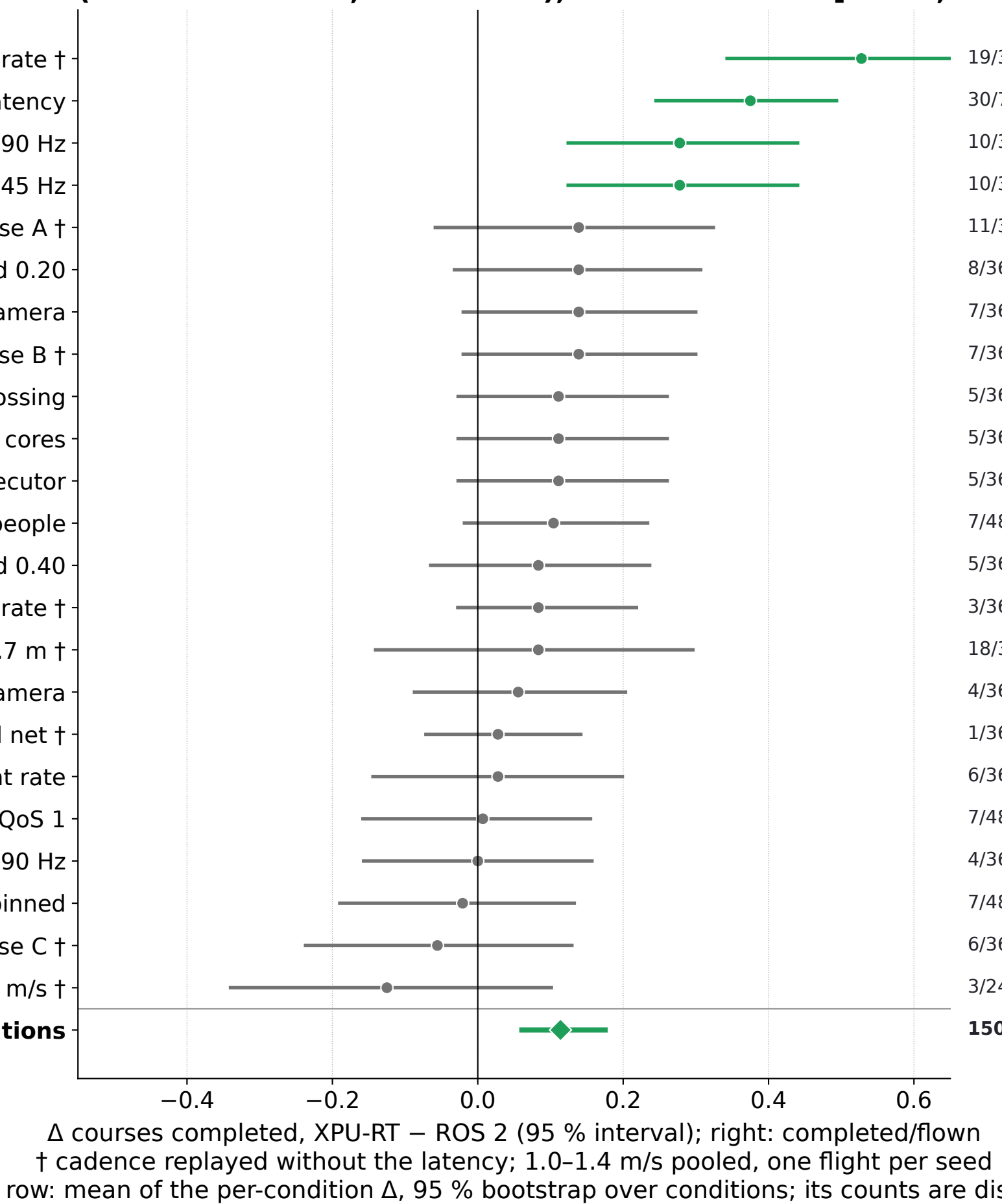
Where flights end (1.7 m scene): median 13.5 vs 5.7 m (2.4 m: 9.3 vs 5.0)



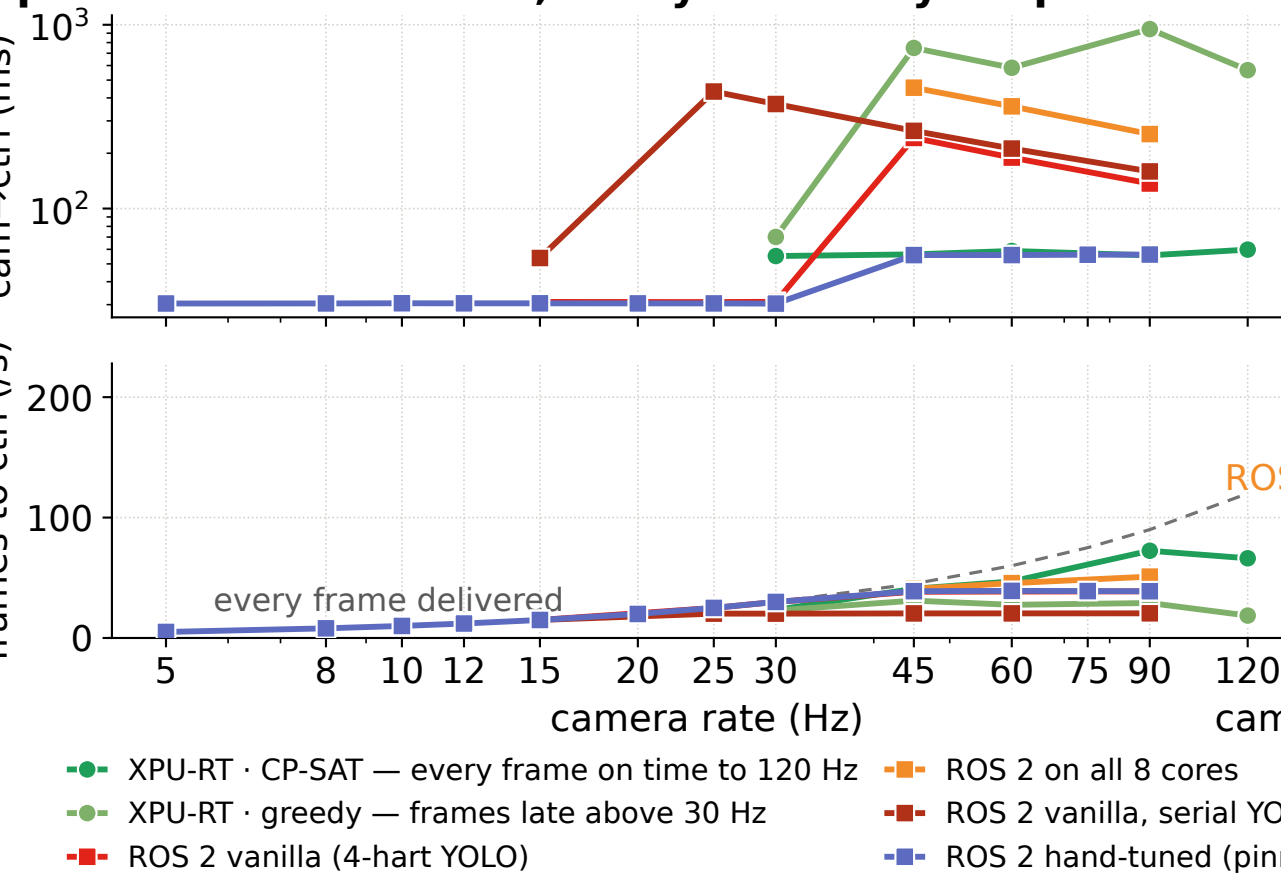
Mechanism: moment (measured), power (model)



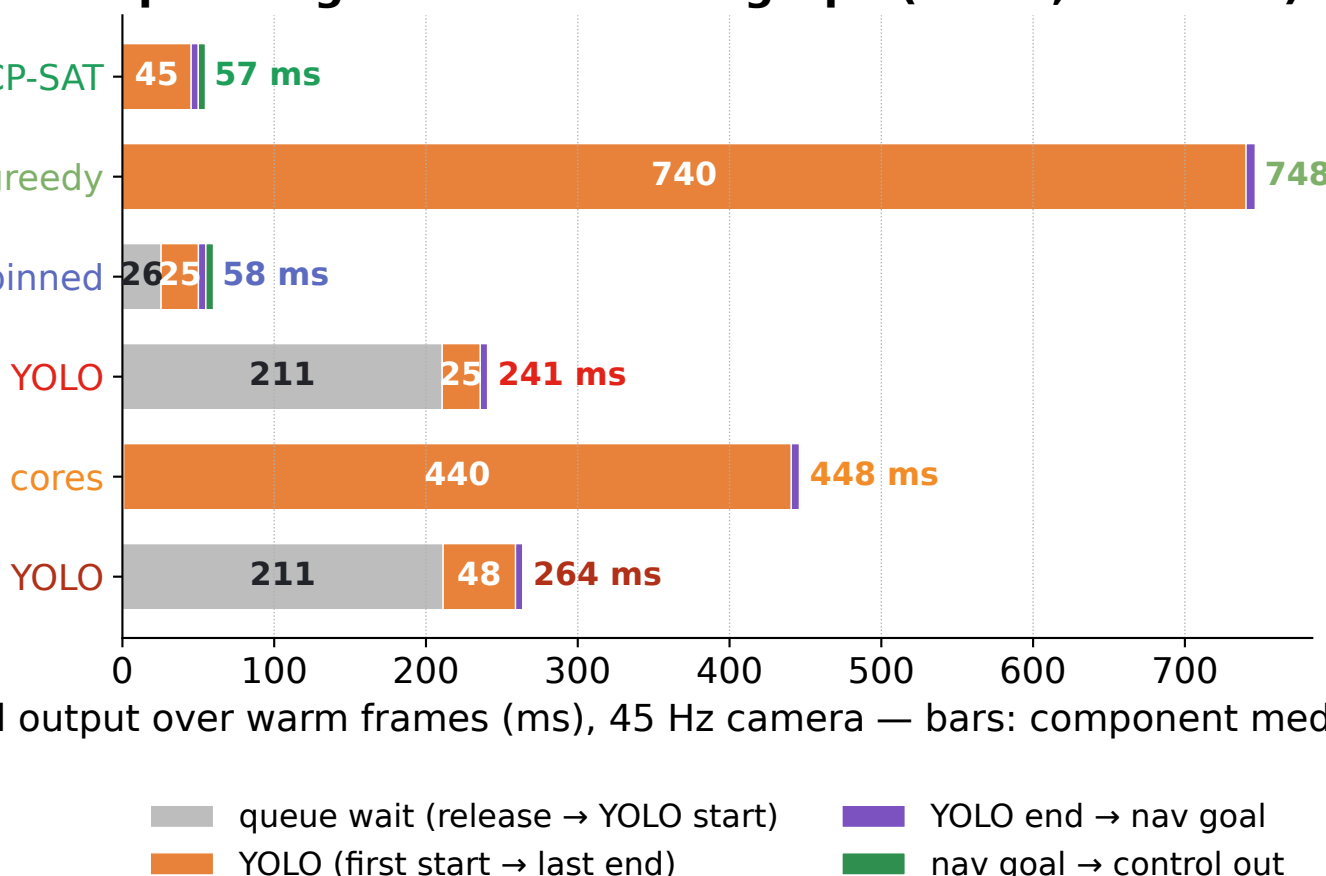
Every condition flown by both arms: XPU-RT ahead in 19 of 23 (clear of zero in 4, behind in 0); mean Δ = +0.11 [+0.06, +0.18], bootstrap over conditions



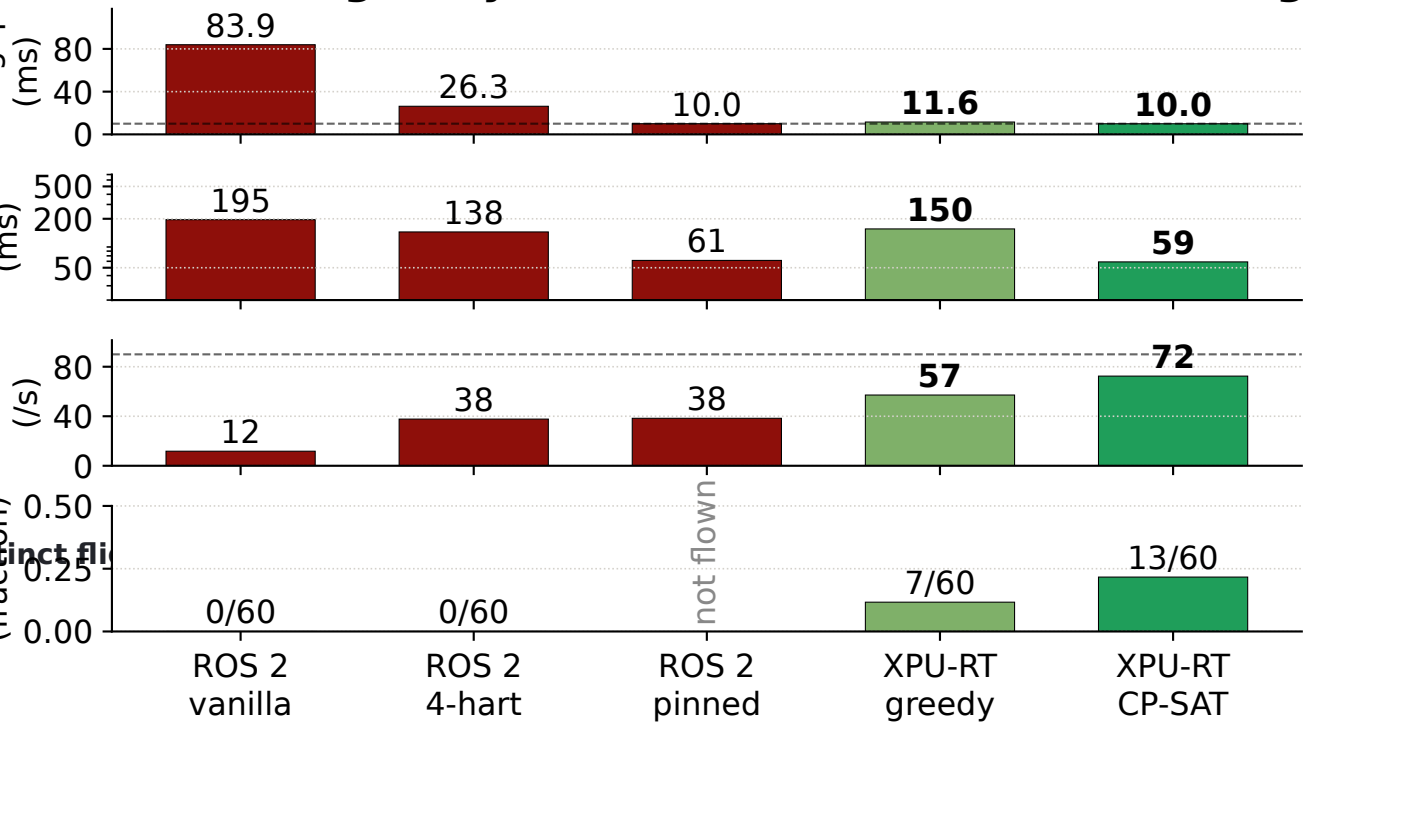
Why, on the board — camera rate: every frame on time to 120 Hz; every ROS 2 layout plateaus



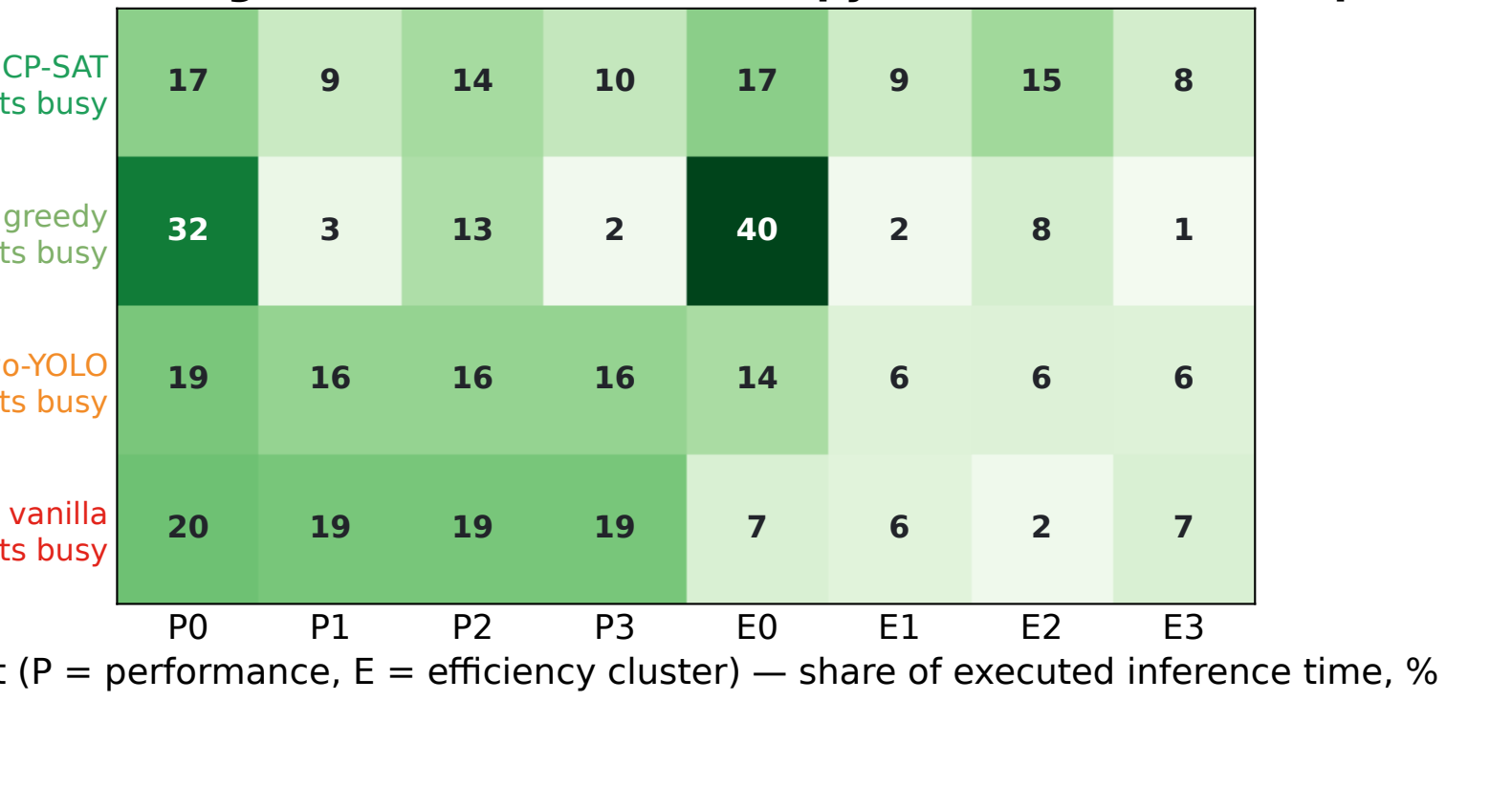
Where the latency goes: compute for the schedule, queueing for the callback graph (45 Hz, medians)



Added load: two more networks on the same harts — CP-SAT > greedy > ROS 2, on the board and in flight



Where the work lands: the schedule spreads the kernels over all eight harts; ROS 2 can occupy them too, and still queues



The schedules themselves — Onboard K1, measured, 45 Hz camera — CP-SAT: 57 ms, every frame on time, control 10.0 ms · greedy: 748 ms, 44/44 late, control 13.2 ms ROS2 8-core: 447 ms, control 16.6 ms · ROS2 vanilla: 242 ms, control 25.6 ms

